Counting

Rule of sum:

If a sequence of tasks $T_1, T_2, \ldots, T_m$ can be done in $w_1, \ldots, w_m$ ways resp. (none of these can be performed simultaneously), then the # of ways to do one of these tasks is $w_1 + w_2 + w_3 + \ldots + w_m$.

e.g. There are 4 types of cookies and 3 types of cakes. In how many ways can we select 1 item to eat?

→ T_1: cookie (4) T_2: cake (3)

$4 + 3 = 7$

Rule of Product:

If a sequence of tasks $T_1, T_2, \ldots, T_m$ can be done in $w_1, w_2, \ldots, w_m$ ways resp. (tasks are done sequentially) (T_1 and T_2 and T_3 and ...)
then the # of ways to perform all the tasks is $w_1 \times w_2 \times w_3 \times \ldots \times w_m$.

e.g. 4 types of cookies, 3 cake types.
of ways to select 1 cookie & 1 cake to eat?

→ $4 \times 3 = 12$

e.g. (X) —3 buses / 2 autos— (Y) —4 buses / 5 autos— (Z)
home school

How many ways to reach school?

→ $(3+2) \times (4+5) = 45$

T_1: X to Y T_2: Y to Z

T_1 and T_2

3 buses or 2 autos → 4 buses or 5 autos

Q. How many divisors of 100 are there?

$100 = 2^2 \times 5^2$

$(3 \times 3) = 9$

2^2: 0, 1, 2 — 3 ways

5^2: 3 ways

Q. 360?

→ $2^3 \times 3^2 \times 5$

of divisors $= 4 \times 3 \times 2 = 24$

In general,

If the prime factorizn of n is $p_1^{e_1} \cdot p_2^{e_2} \cdots p_k^{e_k}$

where p_i's are distinct primes

then # of divisors of n,

$$d(n) = (1+e_1) \cdot (1+e_2) \cdots (1+e_k)$$

Q. What is the sum of all divisors of n?

$$n = p_1^{e_1} \cdot p_2^{e_2} \cdots p_k^{e_k} \quad \text{(prime factorizan)}$$

→ Say $n = 40$. A factor of 40 would be

$40 = 2^3 \times 5$

$$x = 2^a 5^b$$

$$0 \le a \le 3 \qquad 0 \le b \le 1$$



sum of divisors $= 2^0 5^0 + 2^0 5^1 + 2^1 5^0 + 2^1 5^1 + 2^2 5^0$
$+ 2^2 5^1 + 2^3 5^0 + 2^3 5^1$

$= (2^0 + 2^1 + 2^2 + 2^3)(5^0 + 5^1)$

$= \frac{2^0(1-2^4)}{(1-2)} \times \frac{5^0(5^2-1)}{(5-1)}$

$= 15 \times 6 = 90$

So for $p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$

$$\text{Sum of divisors} = \prod_{i=1}^{k} \frac{p_i^{e_i+1} - 1}{p_i - 1}$$

Factorial

$n!$ = # of ways to arrange n different objects in a row.



ABC
ACB
BAC
BCA
CAB
CBA

$3! = 6$

For n objects

row has n places $\underline{\quad}\ \underline{\quad}\ \underline{\quad}\ \underline{\quad}\ \underline{\quad}\ \ldots$

↑ we have n choices

when we put 1, we have $n-1$ objects left

$\underline{\cdot}\ \underline{\quad}\ \underline{\quad}\ \underline{\quad}\ \underline{\quad}$

↑ n ways ↑ $n-1$ ways

$n \times (n-1) \times (n-2) \times \ldots \times 1$

T: arrange n objects

- T_1: put an item in the 1st place
- T_2: arrange remaining $(n-1)$ items.

} T_1 and T_2

$T_2 \rightarrow (n-1)!$

$n! = n \times (n-1)!$

$1! = 1, \quad 0! = 1$

Q. # of factors of 9! = ?

→ $9! = 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$

$= 3^2 \cdot 2^3 \cdot 2 \cdot 3 \cdot 2^2$

$= 2^7 \cdot 3^4 \cdot 5^1 \cdot 7^1$

$8 \times 5 \times 2 \times 2 = 160.$ ← # of factors

Q. What is the # of zeros at the end of 50!

→ 50! will be $2^n \times 5^n \times m$ where $n \geq m$.

So we just have to find n.

$m \times 5 \times 10 \times 15 \times 20 \times 25 \times 30 \times 35 \times 40 \times 45 \times 50$

$5 \quad 5 \quad 5 \quad 5\times5 \quad 5 \quad 5 \quad 5 \quad 5 \quad 5\times5$

12 times 5. 2 is obv. more frequent.

Those will multiply to give 10's at end.

So 12 zeros at the end.

Or

$\left\lfloor \frac{50}{5} \right\rfloor + \left\lfloor \frac{50}{5^2} \right\rfloor + \left\lfloor \frac{50}{5^3} \right\rfloor + \cdots$

$= 10 + 2 + 0 + 0 + \cdots = 12$

Permutations: A permutation of a set is an arrangement of its members into a seq. or a linear order.

of arrangements of r items from n objects is called a partial permutations. [No repetitions]

↳ ${}^nP_r$, read as n permute r.

$$ {}^nP_r = \frac{n!}{(n-r)!} $$

---

n objects, r places

choices → $\underline{n}$ $\underline{n-1}$ $\underline{n-2}$ ... $\underline{n-r+1}$

1st 2nd 3rd ... rth

$$ {}^nP_r = n \cdot (n-1) \cdot (n-2) \cdots (n-r+1) $$

$$ = n \cdot (n-1) \cdot (n-2) \cdots (n-r+1) \times \frac{(n-r)(n-r-1)\cdots 1}{(n-r)(n-r-1)\cdots 1} $$

$$ {}^nP_r = \frac{n!}{(n-r)!} $$

$$ {}^nP_r = \frac{n!}{(n-r)!} \qquad {}^7P_3 = 7 \cdot 6 \cdot 5 $$

e.g. # of 3-letter words (with/without meaning) which can be formed from the word "PARTY" without any repetition?

→ 

$5 \cdot 4 \cdot 3 = 60$ ${}^5P_3$

e.g. # of ways in which the word "CLIMATE" can be arranged? → 7!

e.g. # of ways in which the word "BEAKER" can be arranged?

→ $\frac{6!}{2!}$

AAB → say A_1 A_2 B

$A_1 A_2 B$ → same AAB

$A_1 B A_2$, $A_2 A_1 B$ → same ABA

$A_2 B A_1$

$B A_1 A_2$, $B A_2 A_1$ → same BAA

A is repeated 2 times so every permutation will repeat 2! times. As the # of ways in which the A's arrange themselves would be 1 instead of 2!

AABBBCD → $\frac{7!}{2!3!}$

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e.g. # of permutations of the word CLIMATE s.t. all the consonants occur together.

→ [CLMT] I A E $4! \times 4!$

e.g. # of ways to arrange the letters of the word CHEETAH?

→ $\frac{7!}{2!2!}$

Q. CHEETAH : vowels come together.

[EEA] CHTH $\frac{5!}{2!} \times \frac{3!}{2!}$

Q. All vowels are together, all consonants are together.

→ [EEA] [CHTH] $2! \cdot \frac{3!}{2!} \cdot \frac{4!}{2!}$

Q. vowels and consonants on alternate places.

→

C E H E T A H

C V C V C V C



$\frac{3!}{2!} \times \frac{4!}{2!}$

EEA CHTH

3 vow. 4 cons.

Combinations: # of possible selections in a collection of items where the order of selection is NOT important.

e.g. # of ways to select 2 items out of 3.
→ 123 → 12 or 23 or 13. 3

$$^nC_r = \frac{^nP_r}{r!} = \frac{n!}{(n-r)!\,r!}$$

$$^7C_3 = \frac{7 \cdot 6 \cdot 5}{3 \cdot 2 \cdot 1}$$

Axioms:

Combination as recurrence:

$$^nC_r = {}^{n-1}C_r + {}^{n-1}C_{r-1}$$ Pascal's Identity

Intuitive proof:
say $n=6$, $r=3$. we want to find $^6C_3$.
we've got 6 items. consider the 1st.
we could take it or leave it.
As of the remaining, we've got to select either 3 or 2, depending on 1st choice.

$$^6C_3 = {}^5C_2 + {}^5C_3$$
1 (1st item chosen) ↔ 1st item left out.

(leave or take)

Ily, $^nC_r = {}^{n-1}C_r + {}^{n-1}C_{r-1}$

Proof: $^{n-1}C_r + {}^{n-1}C_{r-1}$

$$= \frac{(n-1)!}{r!\,(n-1-r)!} + \frac{(n-1)!}{(r-1)!\,(n-r)!}$$

$$= \frac{(n-1)!}{(r-1)! \cdot (n-r-1)!} \left[\frac{1}{r} + \frac{1}{n-r} \right]$$

$$= \frac{n \cdot (n-1)!}{r(r-1)!\ (n-r)(n-r-1)!}$$

$$= \frac{n!}{r!\ (n-r)!} = {}^{n}C_{r}$$

Pascal's Δ :

$${}^{0}C_{0}$$

$${}^{1}C_{0}\ \ {}^{1}C_{1}$$

$${}^{2}C_{0}\ \ {}^{2}C_{1}\ \ {}^{2}C_{2}$$

$${}^{3}C_{0}\ \ {}^{3}C_{1}\ \ {}^{3}C_{2}\ \ {}^{3}C_{3}$$

```
          1
        1   1
      1   2   1
    1   3   3   1
  1   4   6   4   1
```

Any ${}^{n}C_{r}$ can be calculated in $O(1)$ provided we have ${}^{n-1}C_{r}$ and ${}^{n-1}C_{r-1}$. (DP)

${}^{n}C_{K} = {}^{n}C_{n-K}$ K selection = K rejection

Q. # of matches played = ?

Facts:

- → 10 teams
- → each team plays with every other team once in group stage.
- → 2 semifinals
- → 1 final

→ ${}^{10}C_{2} = \frac{10 \cdot 9}{2 \cdot 1} = 45$ $\quad 45 + 2 + 1 = 48$

Q. A ques^n paper consists of 16 ques^{n}s with 8 questions each in sets A & B.
The students have to ans. 5 ques^{n}s selecting atleast 2 ques^{n}s from each set.
In how many ways can a student select 5 ques^{n}s to answer?

→ $^8C_2 \cdot {}^8C_3 + {}^8C_3 {}^8C_2$

Q. # of diagonals in an n-sided polygon?

→ $^nC_2 - n$

Perm. vs comb.

↓ arrangements, order matters

↘ selection, order does not matter

Combination with Repetition:

Assume that we have a set A with n elements. Any selection of r objs from A, where each object can be selected more than once is called a comb^n of n objs taken r at a time with repetition.

Q. # of ways of selecting 3 letters from $\{a, b, c, d\}$ with repetition allowed?

→

			20 ways.
aaa	acd	ccc	
aab	add	ccd	
aac	bbb	cdd	
aad	bbc	ddd	
abb	bbd		
abc	bcc		
abd	bdd		
acc	bcd		

We'll essentially take some a's, some b's, some c's, some d's.

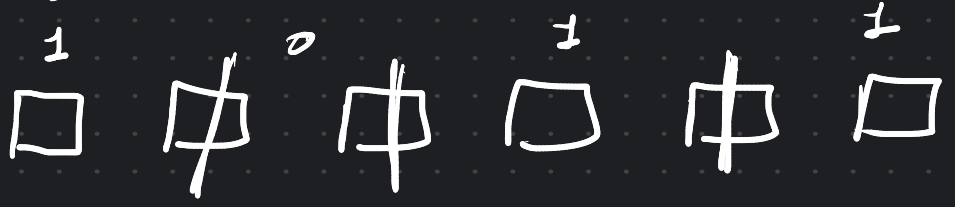
$$n_a + n_b + n_c + n_d = 3$$

Restriction is that $0 \leq n_a \leq 3$, same for others.

→ We need # of distinct sol^{n}s to this.

$$n_a + n_b + n_c + n_d = 3$$

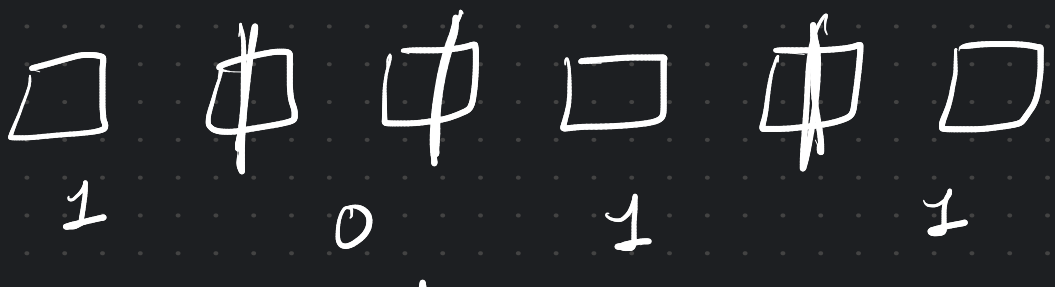
Insert 3 sticks into 6 boxes.



Gaps betn the sticks will be a unique sol^n to the eq^n. (empty boxes)



bbc



acd

So # of ways to put 3 sticks in 6 boxes is same as our original problem.

$= {}^6C_3 = 20$

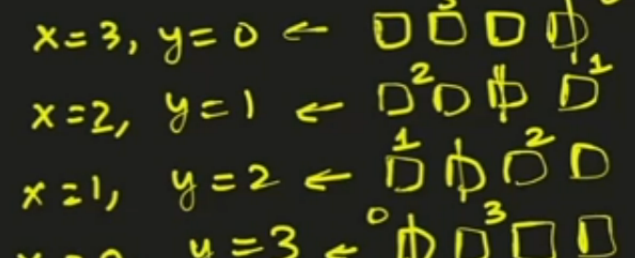
∴ same as original problem.

$x + y = 3$, x, y are non-negative integers.

sol^n's are $(0,3), (1,2), (2,1), (3,0)$. 4.

same as 4 boxes, 1 stick.

of ways $= {}^4C_1 = 4$



same as having 2 types of objs & having to select 3 items with repetition.

of sticks = # of partitions − 1

of boxes = remaining boxes + boxes with sticks

ought to be the same

---

e.g. Say there are 3 flavors of juices. 10 friends want to buy juice. How many variations of the order are possible?

→ ABC → variations will be like 3A, 2B, 5C.

$a + b + c = 10$

3 partitions: so 2 sticks.

2 + remaining boxes = total boxes

2 + 10 = boxes

12 boxes.

One instance:



(4A, 2B, 4C)

of total ways = $_{12}C_2 = \frac{12 \cdot 11}{2 \cdot 1} = 66$

Say n diff. objects. r to be selected.

$$x_1 + x_2 + \dots + x_n = r$$

of sticks $= (n-1)$.

of boxes $= n + r - 1$

of ways to choose boxes in which to put sticks $= {}^{n+r-1}C_{n-1}$

$I_1 \quad I_2 \quad I_3 \cdots \quad I_n$ — n objects

$$\begin{cases} x_1 + x_2 + x_3 + \cdots + x_n = r & r \text{ items} \\ \forall\, x_i \geq 0 \end{cases}$$

n partitions → # sticks $= n-1$

boxes $= n-1+r$

$$\boxed{{}^{n+r-1}C_{n-1}} = \boxed{{}^{n+r-1}C_{r}}$$

(k-sol^n = k-soln.)

e.g. $x + y + z = 12$

$n = 3$

$r = 12$

${}^{n+r-1}C_r$

$= {}^{3+12-1}C_{12} = {}^{16}C_{12}$

Q: In how many ways can 50 be represented as sum of 3 whole numbers?

≡ In how many ways can we select 50 items out of 3 distinct objects with repetitic

→ $a+b+c=50, \quad a,b,c \geq 0$

of partitions = 3, # of sticks = 2

of boxes = 2 + 50 = 52

$^{52}C_2$

=

Q2: In how many ways can 50 be represented as sum of 3 natural numbers?

≡ In how many ways can we select 50 items out of 3 distinct objects with rep allowed given that we select at least one item of each type?

→ $a+b+c=50, \quad a,b,c \geq 1$

⇒ $a+b+c \geq 47 \quad a,b,c \geq 0$

$n=3 \quad r=47$

$^{n+r-1}C_r = {}^{50}C_{47}$

Q.3. In how many ways can 150 identical candies be distributed among 6 children?

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 150 \qquad x_i \geq 0$$

$n = 6, \; r = 150$

$${}^{6+150-1}C_{150}$$

Q.4. In how many ways can we fill a bucket with 150 balls of 6 different colors?

→ $c_1 + c_2 + c_3 + c_4 + c_5 + c_6 = 150 \qquad c_i \geq 0$

same as above.

Combination with Repetition:

→ The # of combin's/selec^n's of n objects taken r at a time with repetition.

→ The # of ways r identical objs can be distributed among n distinct objs.

→ The # of non-negative integral soln^s to

$$x_1 + x_2 + \dots + x_n = r$$

↳

$$\binom{n+r-1}{n-1} = \binom{n+r-1}{r}$$

Permutation with repetitions

We have n diff. types of objs. and want to arrange r objs. with repetitions.



$$n \times n \times n \times \cdots \times n = n^r$$

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	without repetition	with repetition
permutation	${}^nP_r$	n^r
combination	${}^nC_r$	$\binom{n+r-1}{r}$

Binomial Theorem:

Binomial Theorem

$(x+y)^2 = (x+y)\cdot(x+y)$

$= xx + xy + yx + yy$

$= x^2 + 2xy + y^2$

1 x^2 + 2 xy + 1 y^2

$(x+y)^3 = (x+y)\cdot(x+y)\cdot(x+y)$

$xxx + xxy + xyx + xyy + yxx + yxy + yyx + yyy$

$= x^3 + 3x^2y + 3xy^2 + y^3$

1 x^3 3 x^2y 3 xy^2 1 y^3

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$$(x+y)^n = (x+y)(x+y)\cdots(x+y)$$

$$= x^n + {}^nC_1 x^{n-1} y + {}^nC_2 x^{n-2} y^2 + \cdots + {}^nC_n y^n$$

$$(x+y)^n = \sum_{k=0}^{n} {}^nC_k \, x^{n-k} y^k$$

For $x, y = 1$:

$${}^nC_0 + {}^nC_1 + \cdots + {}^nC_n = 2^n$$

multinomial theorem

$(x+y+z)^2 = (x+y+z)(x+y+z)$

sum of powers $x^r y^s z^t$ will obv. be 3 for all terms

$= x^2 + y^2 + z^2 + 2xy + 2yz + 2xz$

$(x+y+z)^3$ we'll have terms $x^a y^b z^c$

$a, b, c \geq 0$

$a+b+c = 3$.

${}^5C_2$ ← soln's.

∴ 10 diff. terms.

$x^3 \quad xy^2 \quad xz^2 \quad xyz \quad x^2y$

$x^2z \quad y^3 \quad z^3 \quad y^2z \quad yz^2$

coeff. of any term:

say x^2y. out of 3 brackets, we choose 2 to get x's. From rest 1, we get a y.

↳ ${}^3C_2 \cdot {}^1C_1$

Consider $(x_1 + x_2 + x_3 + x_4)^{10}$.

Coeff. of $x_1^2 x_2^3 x_3^1 x_4^4$

$$= {}^{10}C_2 \cdot {}^{8}C_3 \cdot {}^{5}C_1 \cdot {}^{4}C_4$$

$$= \frac{10!}{2!\,8!} \cdot \frac{8!}{3!\,5!} \cdot \frac{5!}{1!\,4!} \cdot \frac{4!}{1!\,4!}$$

$$= \frac{10!}{2!} \cdot \frac{1}{3!} \cdot \frac{1}{1!} \cdot \frac{1}{4!}$$

$$= \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5}{2 \cdot 3 \cdot 4}$$

$$= 20 \cdot 63 \cdot 10$$

$$= 12600$$

Do coeff. of $x_1^2 x_2^3 x_3^1 x_4^4$

$$= \frac{10!}{2!\,3!\,1!\,4!} = {}^{10}C_{2,3,1,4}$$

$$= {}^{n}C_{k_1} \cdot {}^{n-k_1}C_{k_2} \cdot {}^{n-k_1-k_2}C_{k_3} \cdots {}^{k_m}C_{k_m}$$

($k_1 \to x_1$, $k_2 \to x_2$, $k_3 \to x_3$, $k_m \to x_m$)

$$= \frac{n!}{(n-k_1)!\,k_1!} \cdot \frac{(n-k_1)!}{(n-k_1-k_2)!\,k_2!} \cdot \frac{(n-k_1-k_2-\cdots-k_{m-1})!}{0!\,k_m!}$$

$$= \boxed{\frac{n!}{k_1!\,k_2!\cdots k_m!}} = {}^{n}C_{k_1,k_2,k_3\cdots k_m}$$

multinomial theorem

The multinomial theorem describes how to expand a power of a sum in terms of powers of the terms in that sum.
It is the generalization of the binomial theorem from binomials to multinomials.

$$(x_1 + x_2 + \cdots + x_m)^n = \sum_{k_1+k_2+\cdots+k_m=n} \binom{n}{k_1, k_2, \ldots k_m} \underbrace{\prod_{t=1}^{m} X_t^{k_t}}_{\downarrow}$$

$$x_1^{k_1} x_2^{k_2} \cdots x_m^{k_m}$$

where $\binom{n}{k_1, k_2, \ldots, k_m} = \dfrac{n!}{k_1!\, k_2! \ldots k_m!}$

Q. sum of all the coeffs. ?

$$(x_1 + x_2 + \cdots + x_m)^n = \sum_{\Sigma k_i = n} \binom{n}{k_1, k_2, \ldots, k_m} x_1^{k_1} x_2^{k_2} \cdots x_m^{k_m}$$

Put $x_1 = x_2 = \cdots = x_m = 1$

$$\underbrace{m^n}_{\uparrow} = \sum_{\Sigma k = n} \binom{n}{k_1, k_2, \ldots, k_m}$$

Q. # of terms in multinomial expansion?

$x_1^{k_1} x_2^{k_2} \cdots x_m^{k_m}$ s.t. $k_1 + k_2 + \cdots + k_m = n$

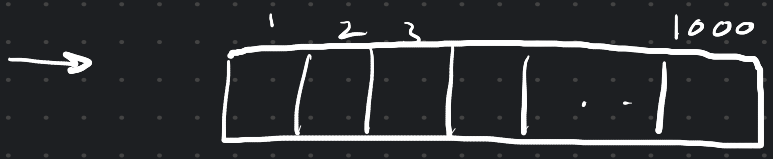
$k_1, k_2, \cdots, k_m \geq 0$

$m-1$ sticks

$n+m-1$ boxes.

$$\binom{n+m-1}{m-1} = \binom{n+m-1}{n}$$

Your mother-in-law buys 1000 small gifts to give to relatives for Christmas. Each of the 1000 things is different.
There are 25 relatives to give gifts to. How many ways are there to distribute the gifts?



$25 \cdot 25 \cdot 25 \cdots 25 = 25^{1000}$

I_1 I_2 I_3 I_4 I_5 5 different objects

$5 \times 4 \times 3$
R_1 R_2 R_3

① distribute 1 gift each to R_1, R_2, R_3, rep NOT allowed.

$5P_3$

② distribute 5 gifts to 3 relatives. (rep NOT allowed)

G_1 G_2 G_3 G_4 G_5

$3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 = 3^5$

$R_1 R_2 R_3$

③ distribute 1 gift each to $R_1 R_2 R_3$, with rep. allowed (assume ∞ supply)

$5 \times 5 \times 5 = 5^3$

R_1 R_2 R_3

④ In how many ways we can distribute s.t. R_1 gets 2 gifts, R_2 gets 2 gifts and R_3 gets 1 gift.

→ ${}^5C_2 \cdot {}^3C_2 \cdot {}^1C_1$ i.e ${}^5C_{2,2,1}$

⑤ 5 identical gifts, distribute them to 3 relatives. How many ways can we do this?

$x_1 + x_2 + x_3 = 5 \qquad x_1, x_2, x_3 \geq 0$

${}^{3+5-1}C_5 = {}^7C_5$

⑥ 5 identical gifts, in how many ways can we distribute s.t. R_1 gets 2 gifts, R_2 gets 2 gifts and R_3 gets 1 gifts.

1

⑦ 5 identical gifts., distribute s.t. each relative gets at least one gift?

→ $x_1 + 1 + x_2 + 1 + x_3 + 1 = 5 \qquad x_1, x_2, x_3 \geq 0$

$x_1 + x_2 + x_3 = 2$

${}^4C_2$

⑧ 5 different items, R_1, R_2, R_3 → s.t. each relative gets at least one gift.

→ solved using inclusion-exclusion.

Q.1 How many positive integers less than or equal to 100 are non-multiples of 2 or 3

→ multiples of 2: $2, 4, 6, 8, \ldots, 100 \to 50$
$1, 2, 3, \ldots, 50$

multiples of 3: $3, 6, 9, \ldots, 99 \to 33$

of 2 and 3: $6, 12, 18, \ldots, 96 \to 16$

$$100 - 50 - 33 + 16$$

Q.2. How many positive integers less than or equal to 100 are non-multiples of 2, 3 or 5?



→ $$100 - \left\lfloor \frac{100}{2} \right\rfloor - \left\lfloor \frac{100}{3} \right\rfloor - \left\lfloor \frac{100}{5} \right\rfloor$$
$$+ \left\lfloor \frac{100}{2 \cdot 3} \right\rfloor + \left\lfloor \frac{100}{2 \cdot 5} \right\rfloor + \left\lfloor \frac{100}{3 \cdot 5} \right\rfloor$$
$$- \left\lfloor \frac{100}{2 \cdot 3 \cdot 5} \right\rfloor$$

LCM(2, 3, 5). Here it comes same as prod, but we want LCM.

Inclusion-Exclusion Principle

In its general form, the principle of inclusion-exclusion states that for finite sets A1, A2, ..., An, we have the following identity:

$$|A_1 \cup A_2 \cup \dots \cup A_n| = |A_1| + |A_2| + \dots + |A_n| - |A_1 \cap A_2| - |A_1 \cap A_3| - \dots - |A_{n-1} \cap A_n| + |A_1 \cap A_2 \cap A_3| + \dots + |A_{n-2} \cap A_{n-1} \cap A_n| - \vdots + (-1)^n |A_1 \cap A_2 \cap \dots \cap A_n|$$

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_{i=1}^{n} |A_i| - \sum_{1 \le i < j \le n} |A_i \cap A_j| + \sum_{1 \le i < j \le k} |A_i \cap A_j \cap A_k| - \dots$$

Q. How many integer solutions does the following system of equations have?

x1 + x2 + x3 = 10, 0 <= x1 <= 4, 0 <= x2 <= 4, 0 <= x3 <= 4

→ we can solve for $x_1 + x_2 + x_3 = 10, \quad x_1, x_2, x_3 \ge 0$ } $^{12}C_2$

we'll have to remove sets S_1, S_2, S_3

S_1: $x_1 > 4$ i.e $x_1 \ge 5$

S_2: $x_2 \ge 5$

S_3: $x_3 \ge 5$

we can only consider one at a time so we'll use the inclusion-exclusion principle.

Solving for $x_1 \ge 5$:

$a + 5 + x_2 + x_3 = 10 \quad a \ge 0, x_2 \ge 0, x_3 \ge 0$

$a + x_2 + x_3 = 5 \quad {}^7C_2$

∴ There are ${}^7C_2$ soln's in S_1. same for S_2 & S_3.

$$\binom{12}{2} - \underbrace{\binom{7}{2} - \binom{7}{2} - \binom{7}{2}}_{\text{we removed common elements in } S_1 \& S_2 \text{ twice. same for } (S_2, S_3) \& (S_1, S_3)} + \underbrace{1}_{S_1 \& S_2} + \underbrace{1}_{S_2 \& S_3} + \underbrace{1}_{S_1 \& S_3} - \underbrace{0}_{S_1 \& S_2 \& S_3}$$

$S_1 \& S_2$: $x_1 \geq 5$, $x_2 \geq 5$

$S_1 \& S_2 \& S_3$: $x_1 + x_2 + x_3 = 10$, $x_1, x_2, x_3 \geq 5$

$$= \frac{12 \cdot 11}{2 \cdot 1} - \frac{7 \cdot 6}{2} \cdot 3 + 3 = 66 - 63 + 3 = \underline{6}$$

$a+5 + b+5 + x_3 = 10$

$a + b + x_3 = 0$, $a, b, x_3 \geq 0$

${}^2C_2 = 1$

Q. x + y = 10, y >= 0, 0 <= x <= 6. # of solutions ?

→

$$a + 7 + y = 10$$

$a + y = 3$, $a, y \geq 0$

${}^4C_1$

$${}^{11}C_2 - \underbrace{\{x \geq 7\}}_{{}^4C_1}$$

Q. x + y = 10, 0 <= x <= 5, 0 <= y <= 5. # of solutions ?

→

$${}^{11}C_2 - \underbrace{{}^5C_1}_{S_1: x \geq 6} - \underbrace{{}^5C_1}_{S_2} + \underbrace{0}_{S_1 \cap S_2: x \geq 6 \text{ and } y \geq 6}$$

S_1: all in which $x \geq 5$, i.e. $x \geq 6$

S_2: all in which $y \geq 5$, i.e. $y \geq 6$

Q. x + y = 15, 0 <= x <= 5, 0 <= y <= 5. # of solutions ?

→ $^{16}C_1 - {}^{10}C_1 - {}^{10}C_1 + {}^{4}C_1 = 0$

$S_1 : x \geq 6$ $S_2 : y \geq 6$ $S_1 \cap S_2$ $x \geq 6$ & $y \geq 6$

Q. How many permutations of 26 small case letters of the English alphabet do not have "cat", "dog", or "mule" as a substring?

→ Total # of permutations = 26!

S_1: "cat" is a substring [cat], b, d, e, ..., z — 24!

S_2: "dog" is a substring 24!

S_3: "mule" is a substring 23!

$S_1 \cap S_2$: "cat" & "dog" are substrings [cat] [dog] b, e, ., z — 22!

$S_1 \cap S_3$: "cat" and "mule" 21!

$S_2 \cap S_3$: "dog" and "mule" 21!

$S_1 \cap S_2 \cap S_3$: "cat", "dog", and "mule" (26-3-3-4+3)! — 19!

of permutations =

$$\underline{26!} - \underline{24!} - \underline{24!} - \underline{23!} + \underline{22!} + \underline{21!} + \underline{21!} - \underline{19!}$$

Q. # of ways in which a, a, b, b, c, c, d, d can be arranged s.t. no two consecutive characters are the same

→ Total # of permutations $= \frac{8!}{2!2!2!2!} = \frac{8!}{(2!)^4}$

S_1: "aa" come together [aa] bbccdd → $\frac{7!}{2!2!2!}$

S_2: "bb" together → $\frac{7!}{2!2!2!} = \frac{7!}{(2!)^3}$

S_3: "cc" together → ~~—~~

S_4: "dd" together → ~~—~~

$S_1 \cap S_2$: [aa] [bb] ccdd $= \frac{6!}{(2!)^2}$

$S_1 \cap S_2 \cap S_3$: [aa] [bb] [cc] dd $= \frac{5!}{2!}$

$S_1 \cap S_2 \cap S_3 \cap S_4$ = [aa] [bb] [cc] [dd] $= 4!$

$$\frac{8!}{(2!)^4} - \left[4 * \frac{7!}{(2!)^3}\right] + \left[6 * \frac{6!}{(2!)^2}\right] - \left[4 * \frac{5!}{2!}\right] + 4!$$

one at a time — 2 at a time — 3 at a time — all together

$= 864$.

of onto functions:

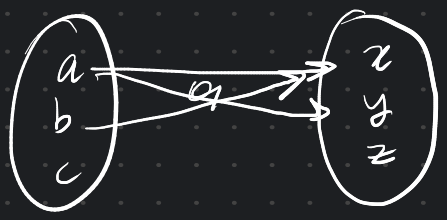
an onto function is one in which range = co-domain

range ⊆ codomain

Q. What is the # of onto functions possible from set A to set B if |A| = n, |B| = m?

onto means there isn't an element in B without a pre-image

→ Say $|A| = |B| = 3$



Total # of functions $= 3^3$.

(a, b, c each have 3 options)

S_1: x has no pre-image 2^3 (a, b, c have 2 choices y, z)

S_2: y has no pre-image 2^3

S_3: z has no pre-image 2^3

$S_1 \cap S_2$: x, y have no pre-image 1^3

$S_2 \cap S_3$: 1

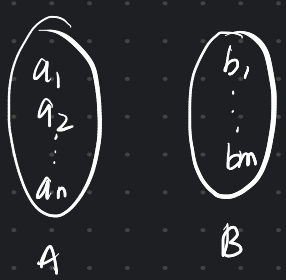
$S_1 \cap S_3$: 1

$S_1 \cap S_2 \cap S_3$: none of them have a pre-image 0 (a, b, c must have a unique mapping)

There should be atleast 1 element to map to.

$$3^3 - 3 \cdot 2^3 + 3 \cdot 1 - 0$$

$$= 6$$



Generalized, |A| = n, |B| = m

$$m^n - \binom{m}{1}(m-1)^n + \binom{m}{2}(m-2)^n - \cdots + (-1)^{m-1}\binom{m}{m-1}1^n + 0$$

m^n: total

$\binom{m}{1}$: one at a time has no pre-img

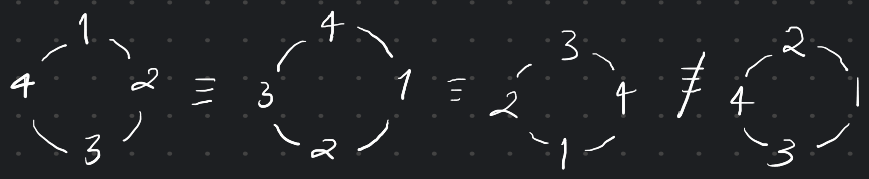
Circular Arrangement:

- permutations, when considered as arrangements, are sometimes referred to as linearly ordered arrangements.
  In these arrangements, there's a first element, a second, and so on.

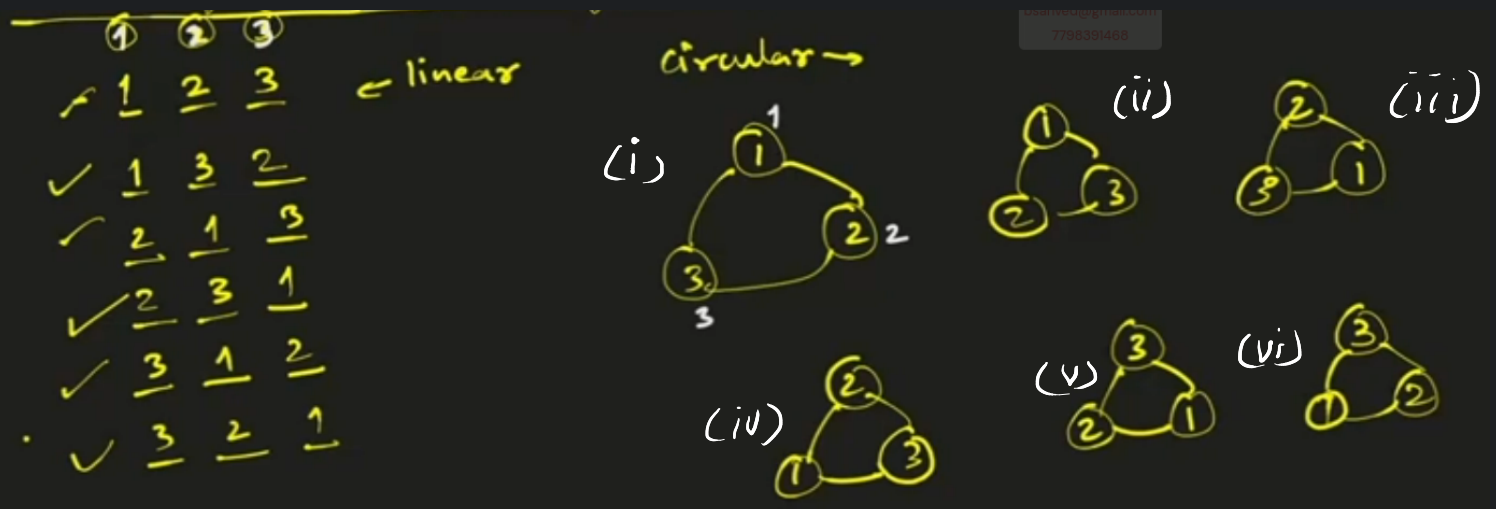
- although if the objects are arranged in a circular manner, this distinguished ordering no longer exists.
  Any element can be the start.



- 2 circular permutations are equivalent if one can be rotated to get the other (i.e. cycled without changing the relative positions of the elements).



of Circular Arrangements:



i, iv, and v are equivalent.
ii, iii, and, vi are equivalent.

From all the above circular permutations,
we can make groups of n (3 here) together as equivalent.
Only 2 are unique.

$$\frac{3!}{3} = 2$$

of circular permutations of a set S with n elements is (n-1)!

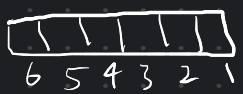
Linear:
1, 2, 3, ..., n

total n! arrangements

Circular:

Setup: 3 couples, (A-B, C-D, E-F) i.e. 6 people are sitting on a table.

Q. # of linear arrangements?
6! = 720



Q. # of circular arrangements?
5! = 120

Q. # of (linear) ways A and B always sit together?
5! * 2! = 240



Q. # of (circular) ways A and B always sit together?
4! * 2! = 48

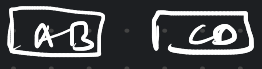
Q. # of (linear) ways no couple sits together?

Total = 6!

S_1: A-B together = 5! 2!

S_2: C-D together = 5! 2!

S_3: E-F together = 5! 2!

S_1 and S_2: 

= 4! 2! 2!

S_2 & S_3 : 4! 2! 2!

S_1 & S_3 : 4! 2! 2!

S_1 & S_2 & S_3 : 3! 2! 2! 2!

of ways = $6! - 3 \cdot 5! 2! + 3 \cdot 4! 2! 2! - 3! 2! 2! 2!$

Q. # of (circular) ways no couple sits together?

$= 5! - 3 \cdot 4! 2! + 3 \cdot 3! 2! 2! - 2! 2! 2! 2!$

$= 32$

Derangements:

- Consider 3 numbered letters 1, 2, 3 to be placed in 3 envelopes numbered 1, 2, 3

- # of ways to place them such that no letter goes in the same numbered envelope? (assume no cover remains empty, i.e. one letter in one envelope)

we could do it using inclusion exclusion:

S_1: 1 is in position 1
S_2: 2 is in position 2
S_3: 3 is in position 3

$3! - 3\times 2! + 3\times 1! - 1$

$2!$: $|S_1|$; $3\times 1!$: 2 at a time; $1!$: $|S_1 \cap S_2|$

$= 6 - 6 + 3 - 1 = 2$

the same can be done for n objects in a row:

S_i: i^{th} is in position 1

$$n! - {}^nC_1 (n-1)! + {}^nC_2 (n-2)! - \cdots + (-1)^n {}^nC_n 1$$

$=$ mains

$$= n! - \frac{n!}{1!(n-1)!}(n-1)! + \frac{n!}{(n-2)!\,2!}(n-2)! - \cdots + (-1)^n \frac{n!}{n!\,n!}\,n!$$

$$= n! - \frac{n!}{1!} + \frac{n!}{2!} - \cdots + (-1)^n \frac{n!}{n!}$$

$$= n!\left[\frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!}\right]$$

$$= n! \sum_{i=0}^{n} \frac{(-1)^i}{i!} = !n = D_n$$

↑ subfactorial

In combinatorial mathematics, a derangement is a permutation of the elements of a set such that no element appears in its original position.

The # of derangements of a set of size n is known as the subfactorial of n
!n
or
the nth de Montmort number, Dn.

$$!n = D_n = n!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \cdots + (-1)^n \frac{1}{n!}\right]$$

$$= n! \sum_{i=0}^{n} \frac{(-1)^i}{i!}$$

Q. # of derangements of 4 objects?

$$!4 = 4!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!}\right]$$

$$= 4!\left[1 - 1 + \frac{1}{2} - \frac{1}{6} + \frac{1}{24}\right]$$

$$= 12 - 4 + 1$$

$$= 9$$

Say n is really big. Factorial grows really fast.

Say $n = 10$

$$!10 = 10!\left[1 - \frac{1}{1} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \underbrace{\frac{1}{5!} + \frac{1}{6!} - \frac{1}{7!} + \frac{1}{8!} \cdots}_{\text{these are really negligible}}\right]$$

$$1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} \cdots \infty = \frac{1}{e}$$

So let us approximate $n \to \infty$

$$\lim_{n\to\infty} \frac{!n}{n!} = \frac{1}{e}$$

$$\Rightarrow !n \approx \frac{n!}{e} \quad \text{(if } n \text{ is large enough)}$$

$\approx 36.8\%$ of $n!$ approx 36.8% arrangements of the total are derangements.

Q. In how many ways can we arrange 1, 2, 3, 4, 5, 6 such that no even integer is in its original position?

→

1 2 3 4 5 6

S_2: 2 appears in its original position. $= 5!$

S_4: 4 in original position $5!$

S_6: 6 in original position $5!$

$S_{2,4} = S_{2,6} = S_{4,6} = 4!$

$S_{2,4,6} = 3!$

$6! - 3 \times 5! + 3 \times 4! - 3!$

$= 720 - 360 + 72 - 6$

$= 426$

Counting: Recap

- Rule of sum: If there are tasks t1, t2, ..., tm tasks that can be done in w1, w2, ..., wm ways respectively, then the # of ways to do one task among all is w1 + w2 + ... + wm.
- Rule of product: # of ways to do all of these tasks one after the other is w1 * w2 * ... * wm

of divisors of a number whose prime factorization is $n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$

$$= (1+e_1)(1+e_2)\cdots(1+e_k)$$

sum of divisors = $$\prod_{i=1}^{k} \frac{p_i^{e_i+1} - 1}{p_i - 1}$$

	without repetition	with repetition
permutations	${}^{n}P_r$	n^r
combinations	${}^{n}C_r$	$x_1 + x_2 + \cdots + x_n = r$, $x_i \geq 0$ $\binom{n+r-1}{n-1} = \binom{n+r-1}{r}$

Binomial Theorem:

$$(x+y)^n = {}^{n}C_0 x^0 y^n + {}^{n}C_1 x^1 y^{n-1} + \ldots + {}^{n}C_n x^n y^0$$

Multinomial Theorem:

Inclusion/ Exclusion:

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum |A_i| - \sum |A_i \cap A_j| + \sum |A_i \cap A_j \cap A_k| - \cdots$$

$$+ (-1)^{n-1} |A_1 \cap A_2 \cap \cdots \cap A_n|$$

of linear arrangements of n items in a row = n!
of circular arrangements of n items = (n-1)!

of derangements of n items

$$= !n = Dn = n!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \cdots + \frac{(-1)^n}{n!}\right]$$

$$!n \approx \frac{n!}{e} = 36.8\% \text{ of } n!$$ for large n values

—x—

eg: If the 13th term of $(x^2 + \frac{2}{x})^n$ is independent of x. find the sum of divisors of n

→ $(x^2 + \frac{2}{x})^n = {}^nC_0 (x^2)^n \left(\frac{2}{x}\right)^0 + {}^nC_1 (x^2)^{n-1} \left(\frac{2}{x}\right)^1$

$+ \cdots + {}^nC_{12} (x^2)^{n-12} \left(\frac{2}{x}\right)^{12}$

↑ 13th term $+ \cdots$

$(x^2)^{n-12} \left(\frac{2}{x}\right)^{12} \Rightarrow x^{2n-24-12}$

$2n - 36 = 0$

$n = 18 = 2^1 \times 3^2$

Sum of divisors of n

$= \frac{2^{1+1} - 1}{2-1} \cdot \frac{3^{2+1} - 1}{3-1} = \frac{3}{1} \cdot \frac{26}{2} = 39$

${}^nC_{12} (x^2)^{12} \left(\frac{2}{x}\right)^{n-12}$ $\qquad 36 - n = 0$

$x^{24 - n + 12}$

$\left(x^2 + \frac{2}{x}\right)^n = {}^nC_0 (x^2)^0 \left(\frac{2}{x}\right)^{n-0} + \cdots$

13th term → ${}^nC_{12} (x^2)^{12} \left(\frac{2}{x}\right)^{n-12}$

$24 - n + 12 = 0 \qquad n \Rightarrow 36$